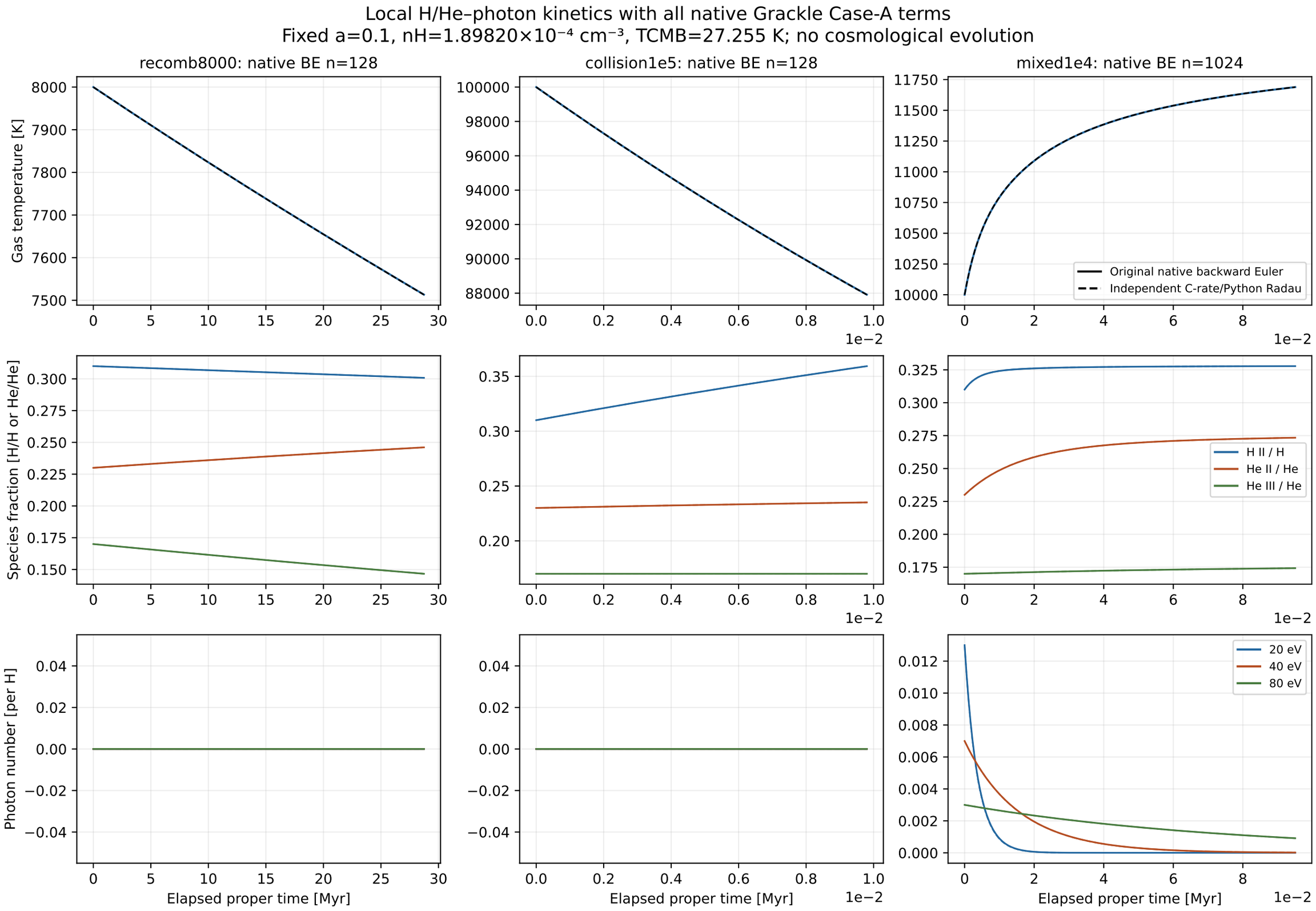
Local H/He–photon kinetics with all native Grackle Case-A terms
Fixed a=0.1, nH=1.89820×10⁻⁴ cm⁻³, TCMB=27.255 K; no cosmological evolution
recomb8000: native BE n=128
collision1e5: native BE n=128
mixed1e4: native BE n=1024
Gas temperature [K]
Species fraction [H/H or He/He]
Photon number [per H]
Elapsed proper time [Myr]
1e−2
Original native backward Euler
Independent C-rate/Python Radau
H II / H
He II / He
He III / He
20 eV
40 eV
80 eV